

VRIJE
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BRUSSEL

Analog Electronics lab session

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Session 3 : Design of differential pairs and of an OTA

In this exercise you will learn ...

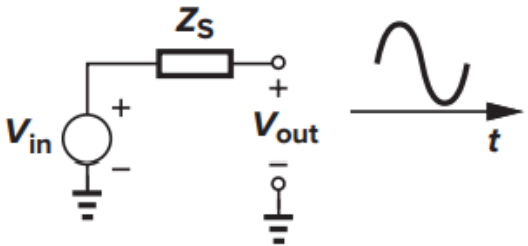
- Fundamentals about differential pairs
- How to "design on paper"
  - Differential pair with resistive load
  - OTA
- Learn to value $\frac{g_m}{I_D}$- and $\frac{g_m}{g_{ds}}$-plots !
  - with those plots one can identify design trade-offs!
- How to code in MATLAB the OTA-design

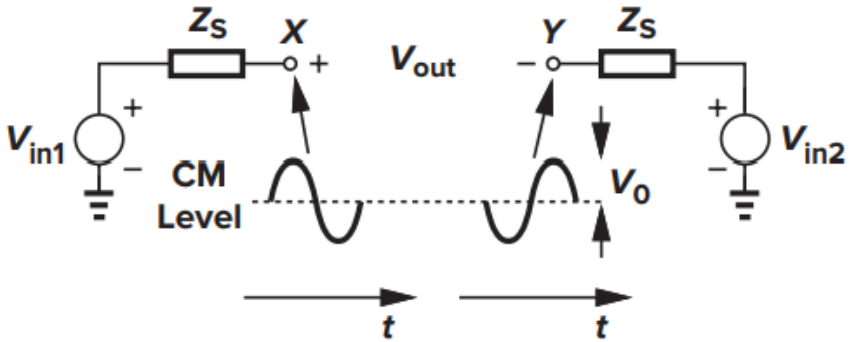
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What is single ended and differential operation?

Single Ended Signal: Signal measured with respect to a fixed potential, usually the ground.

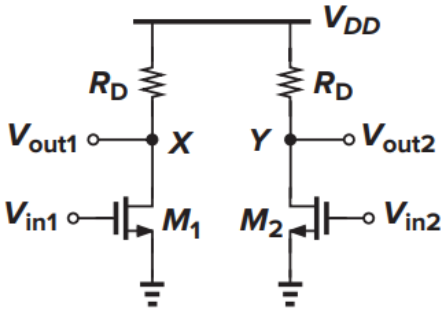
Differential Signal: Signal measured between two nodes that have equal and opposite deviations around a fixed potential.

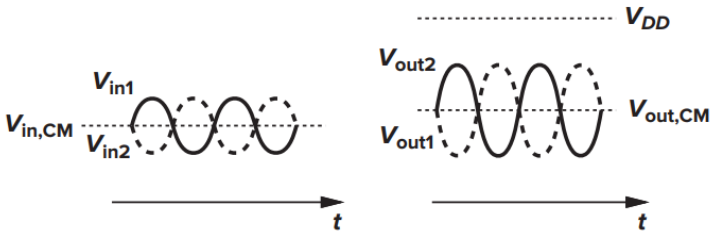




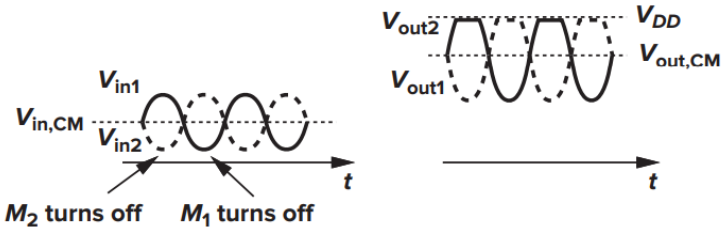
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Principles of a Basic Differential Pair





Normal Operation

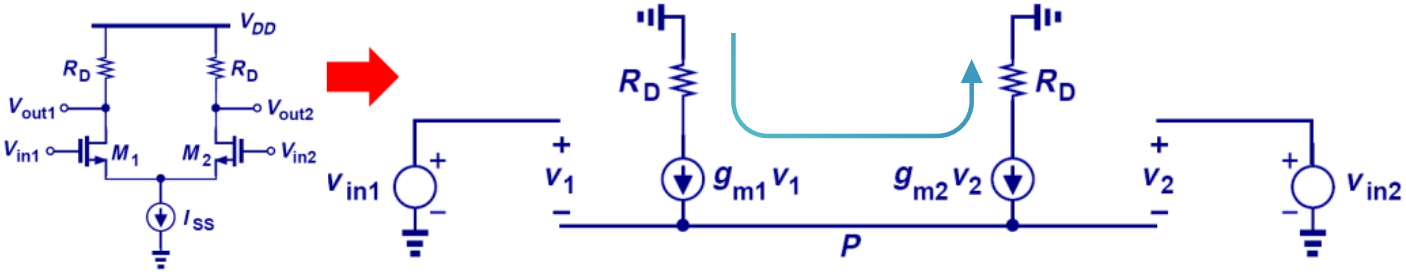


Input CM level is excessively low

Bias currents should have minimal dependence on input CM level!

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Small Signal of a simple Differential Amplifier and Virtual Ground Concept



1. KVL around the input networks
   $v_{in1} - v_1 = v_p = v_{in2} - v_2$

2. KCL at node P
   $gm_1 v_1 + gm_2 v_2 = 0$

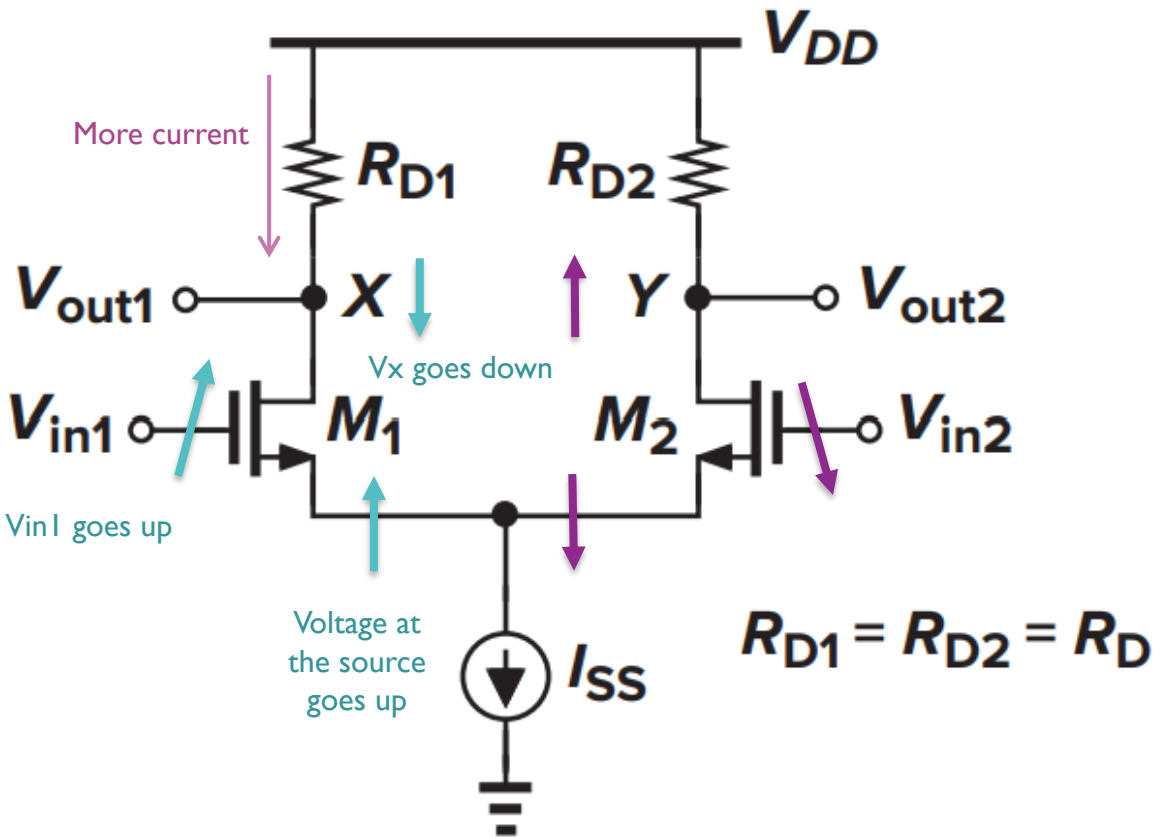
$$-v_1 = v_2$$

$$v_1 = v_{in1}$$

$$v_{in1} - v_1 = v_p = 0$$

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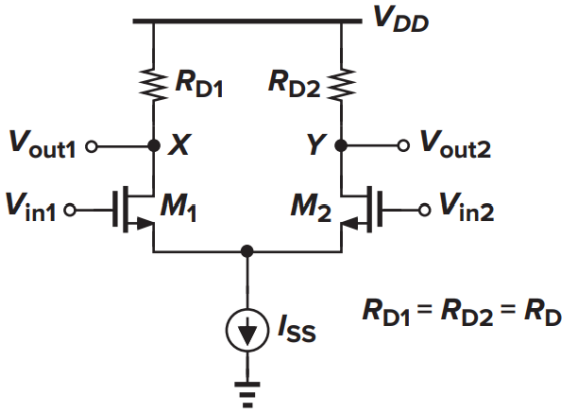
Half Circuit for Differential



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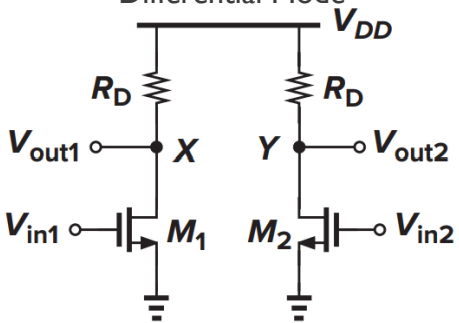
Half Circuit for Differential and Common Mode

For Symmetric Circuits!

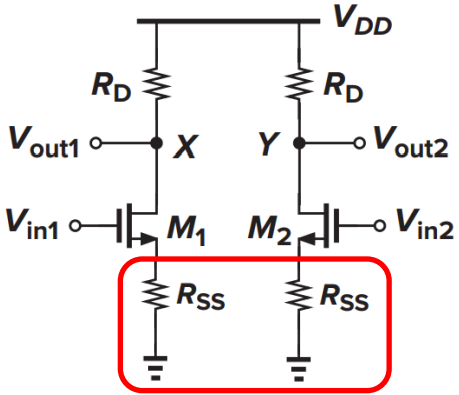


Hint for exercise 1a!!

Differential Mode

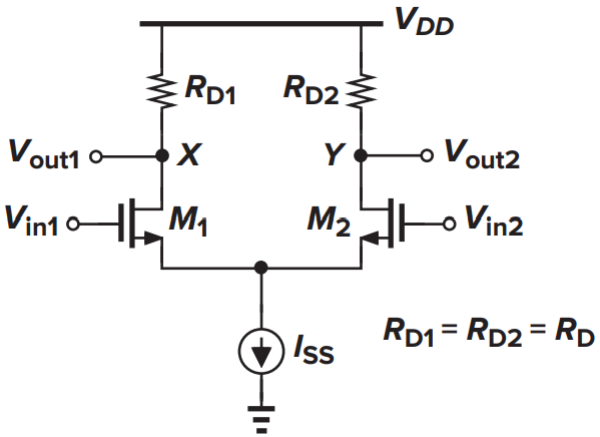


Common Mode



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Large Signal



Think of what happens to nodes X and Y, when $V_{in1} - V_{in2}$ varies from -∞ to + ∞.

When:

- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

Hint for exercise 1b!!

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- Solve exercise 1 a-d
  - ETA: 20 min

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Exercise 1

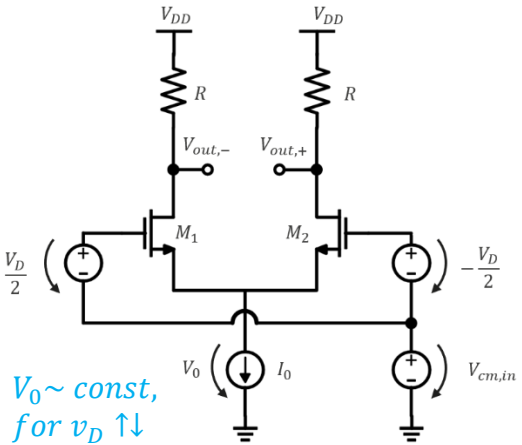
a) Give the equation for $A_{v,-} = \frac{v_{out,-}}{v_D}$, $A_{v,+} = \frac{v_{out,+}}{v_D}$ and $A_{v,diff}$?

a) Remark: M1 = M2

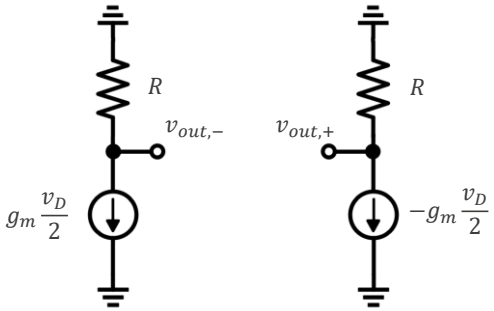
$$A_{V,-} = \frac{v_{out,-}}{v_D} = -\frac{g_m}{2} \cdot R$$

$$A_{V,+} = \frac{v_{out,+}}{v_D} = \frac{g_m}{2} \cdot R$$

$$A_{V,diff} = g_m \cdot R$$



small signal analysis



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Exercise 1.b

b) Draw the voltages V_{out1} and V_{out2} on Fig2 and the difference $V_{out1} - V_{out2}$ on Fig3 as $V_{in1} - V_{in2}$ varies from -∞ to + ∞. Indicate:

i. Upper and lower bound for these voltages
ii. The inversion level of M1 and M2
iii. Small signal voltage gain

Think of what happens to nodes X and Y, when $V_{in1} - V_{in2}$ varies from -∞ to + ∞. Start with the hint!

When:

- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

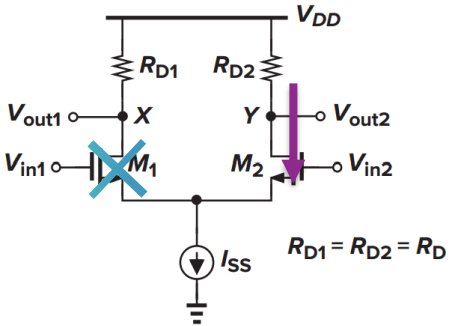
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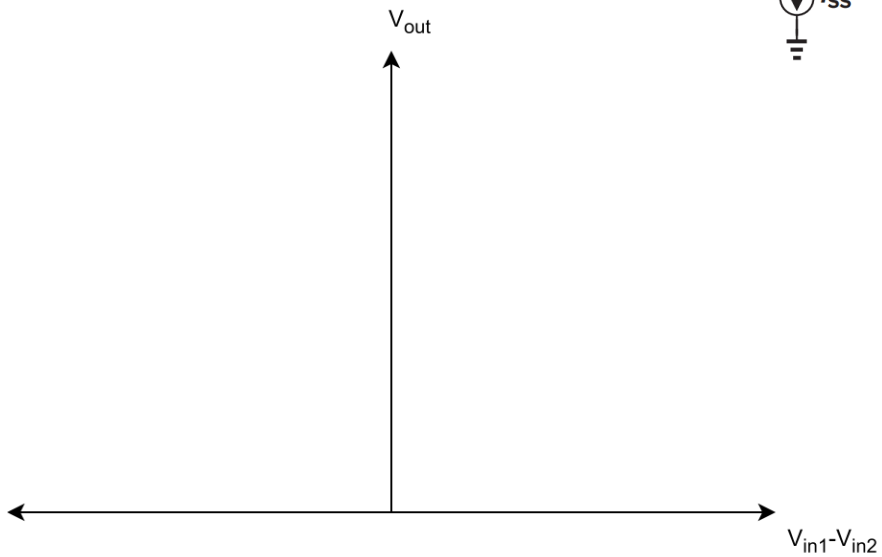
Exercise 1.b

When:

- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

M1 OFF, M2 ON and $I_{D2} = I_{SS}$





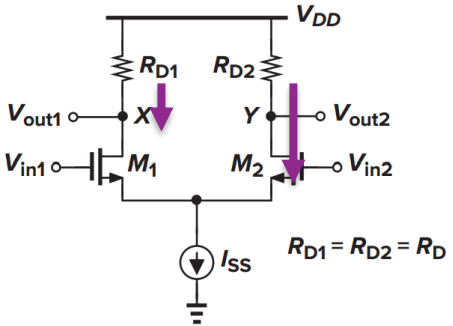
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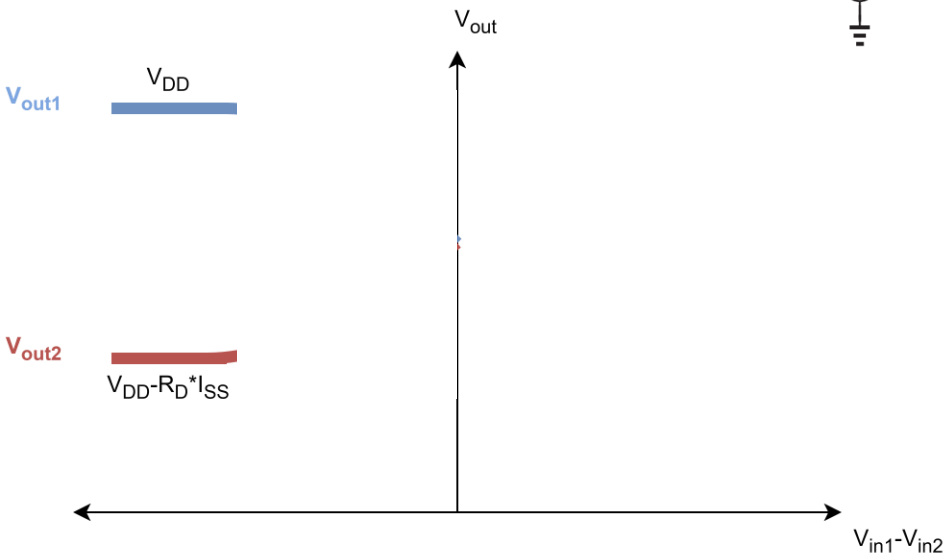
Exercise 1.b

When:

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- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

M1 OFF → ON, M2 ON





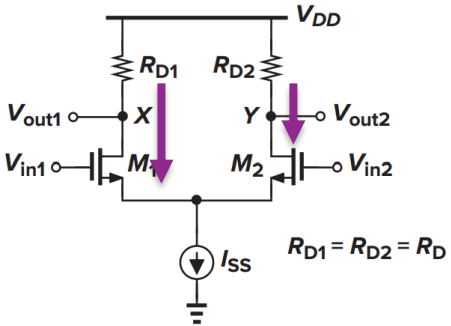
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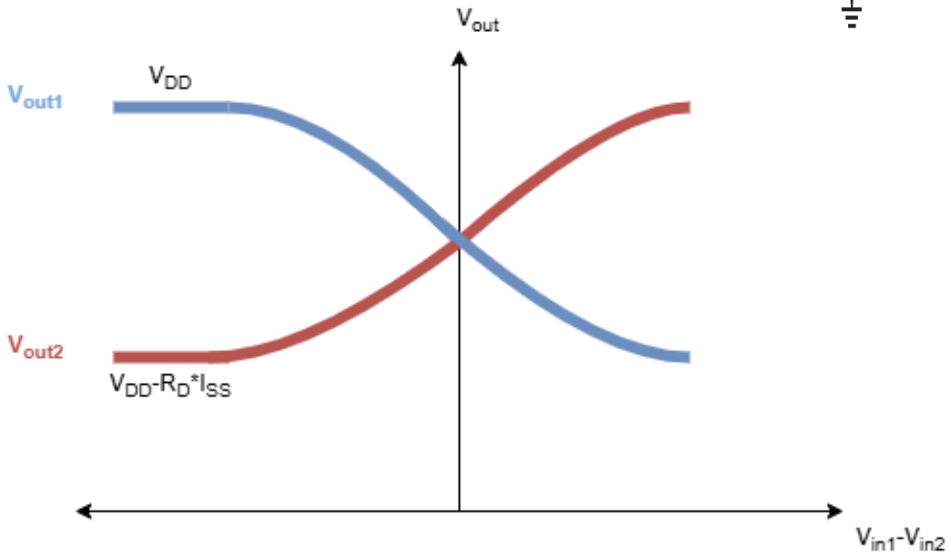
Exercise 1.b

When:

- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

M1 ON, M2 ON → OFF





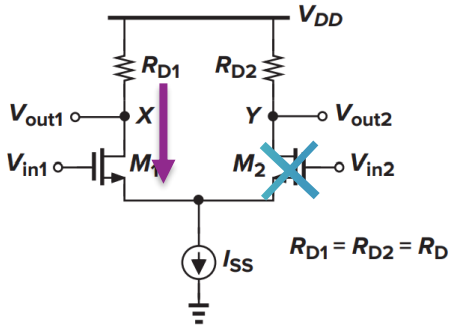
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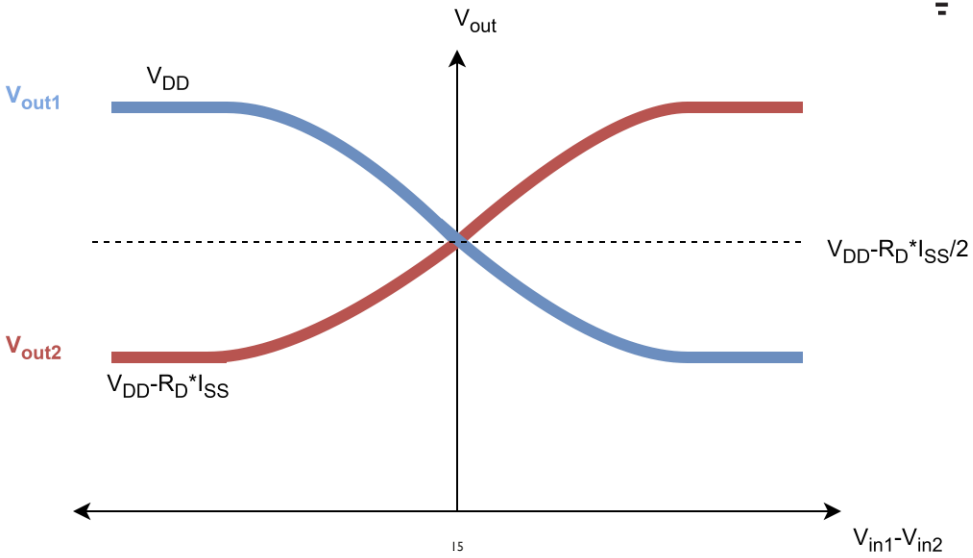
Exercise 1.b

When:

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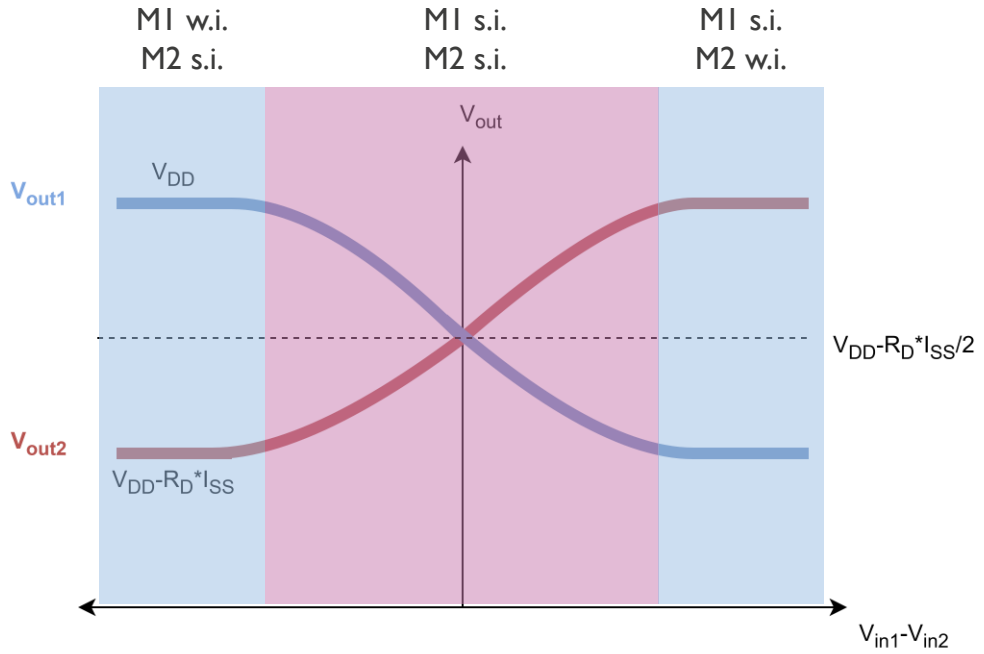
M1 ON, M2 OFF and $I_{D1} = I_{SS}$

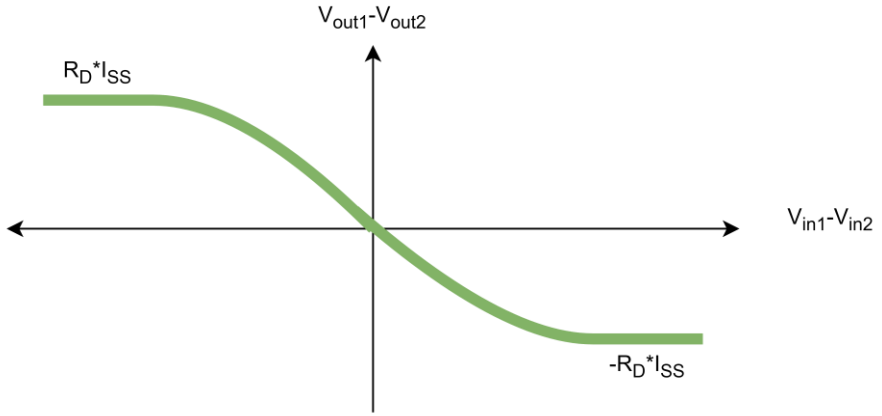




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Exercise 1.b





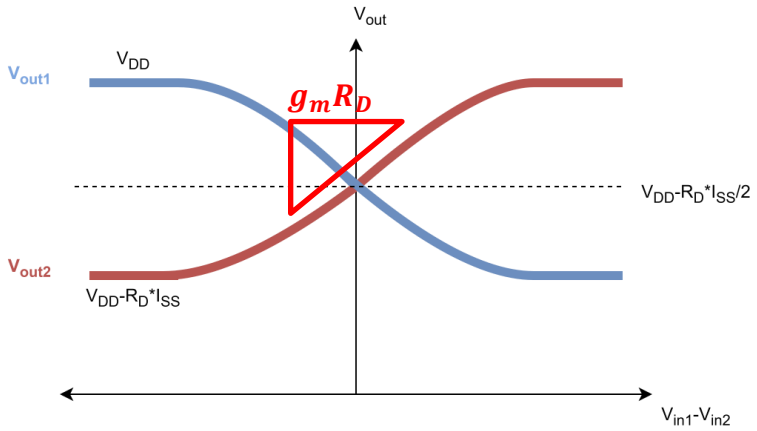
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Exercise 1.b Conclusions

1) The maximum and minimum levels at the output are well-defined.

2) Small signal gain:

$$\frac{V_{out1} - V_{out2}}{V_{in1} - V_{in2}}$$ Maximized for $V_{in1} = V_{in2}$

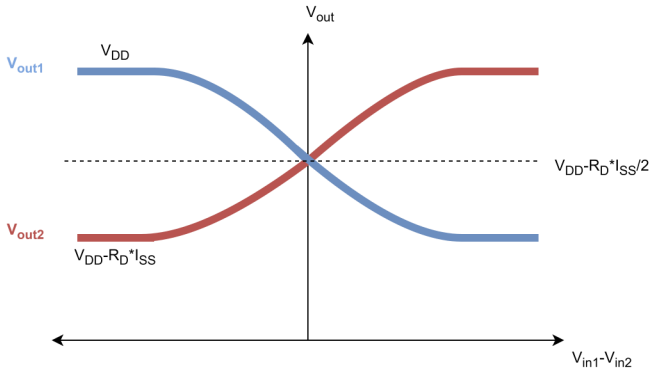


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Exercise 1

c) Give the amount of current, which is flowing through M_1 and M_2, when

(i) $V_D = 0$,
(ii) $V_D << V_{OV}$,
(iii) $V_D >> V_{OV}$.

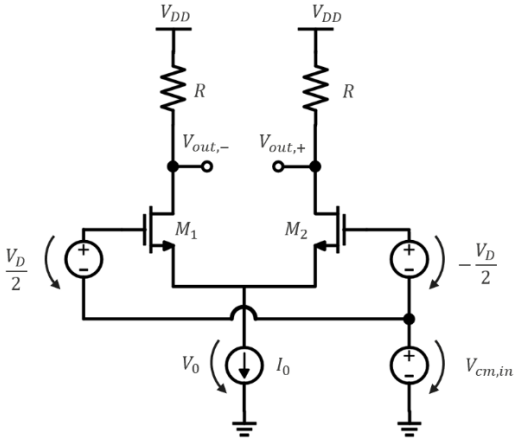


c) $I_{D,M1}$ and $I_{D,M2}$ given as follows

i. $I_{D,M1} = I_{D,M2} = 0.5 \cdot I_0$

ii. $I_{D,M1} = I_0$ and $I_{D,M2} = 0$

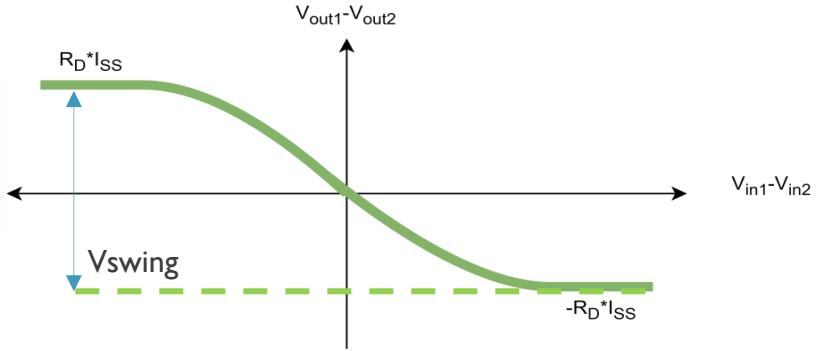
iii. $I_{D,M1} = 0$ and $I_{D,M2} = I_0$



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Exercise 1

d) Based on Fig. 3, derive an equation describing the relationship between V_{swing}, V_{OV} and A_V.
Remark: $V_{swing} = max(V_{out,+} - V_{out,-}) - min(V_{out,+} - V_{out,-})$



d) $\frac{g_{m_{1,2}}}{I_{D_{1,2}}} =$ We know $g_m = \frac{2I_D}{V_{ov}}$ $= \frac{2I_{SS}/2}{V_{OV}} = \frac{2}{(I_{SS}/2)} = \frac{2}{V_{ov}}$

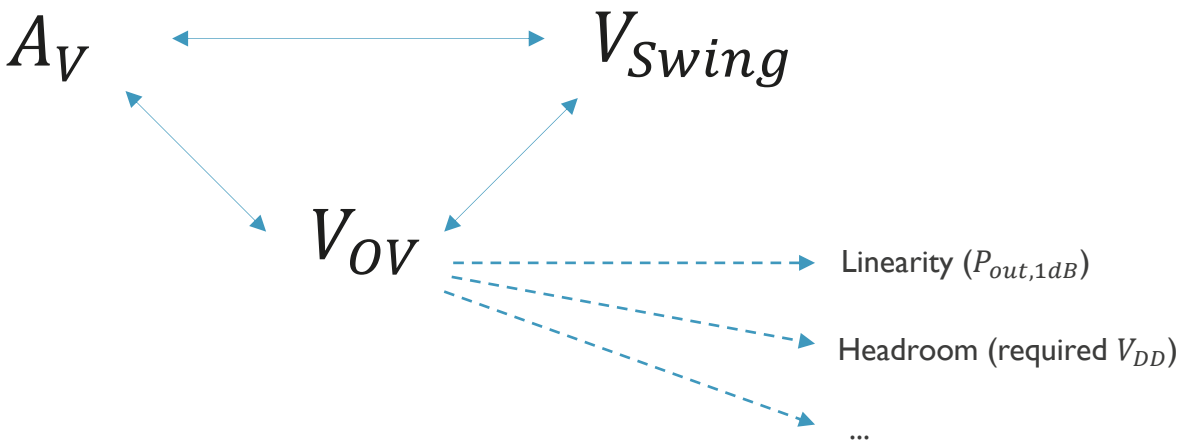
Also, $\frac{g_{m_{1,2}}}{I_{D_{1,2}}} = \frac{g_{m_{1,2}} \cdot R}{(I_{SS}/2) \cdot R} = \frac{2}{V_{OV}}$ ($R = 1$) We know $g_m R = A_{v,diff}$ $\Leftrightarrow 2 \cdot A_{V,diff} = \frac{I_{SS}}{V_{OV}} \cdot 2 \cdot R = \frac{V_{swing}}{V_{OV}}$

$\Leftrightarrow A_{V,diff} = \frac{V_{swing}}{2 \cdot V_{OV}}$

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Summary

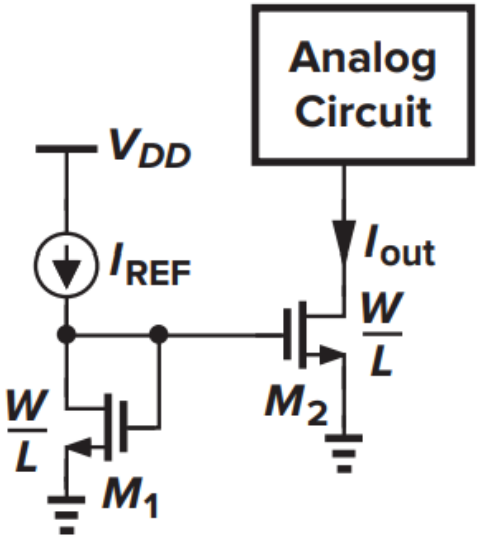
- Waveforms of an ideal differential amplifier has been presented
- (in this exercise) Trade-off triangle of differential amplifiers has been identified
  - ... in reality, one has even an octagon (see literature [1-3])



➢ Now, the differential amplifier is going to be designed

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Exercise 1



Current Mirror Architecture

$$I_{REF} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

Ignore channel length modulation.

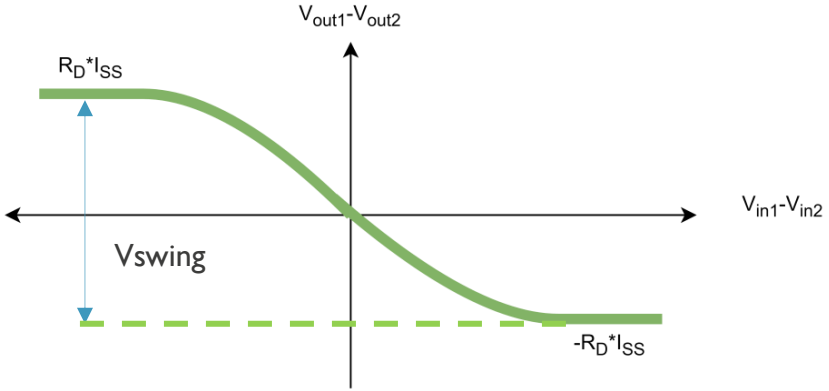
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- Solve exercise 1 e-h (option i&j will be solved together)
  - ETA: 20 min

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Exercise 1

e) Use the plots in option 1.b to give V_{swing} as a function of I_{bias}.

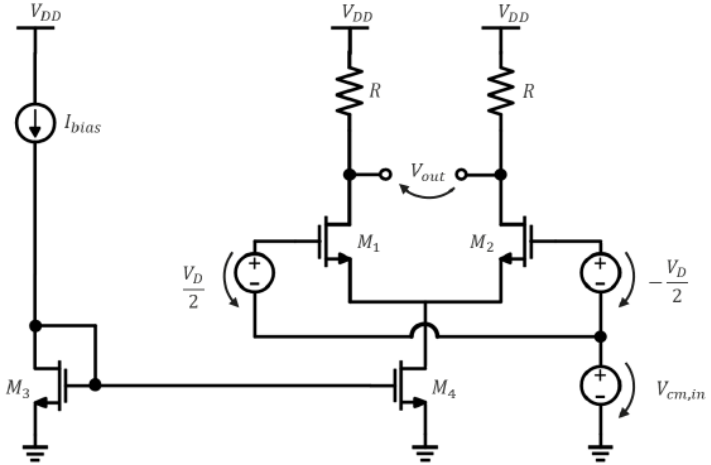


e) $V_{swing} = 2 \cdot R_D \cdot I_{SS}$

current mirror equation (neglecting CLM):

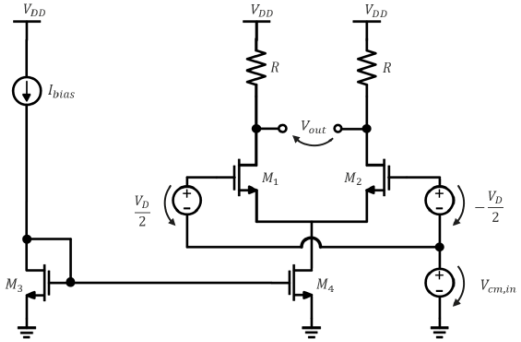
$$\frac{I_{SS}}{I_{bias}} = \frac{W_4}{W_3}$$

$$\Rightarrow \; V_{swing} = 2 \cdot R \cdot I_{bias} \cdot \left(\frac{W_4}{W_3}\right)$$



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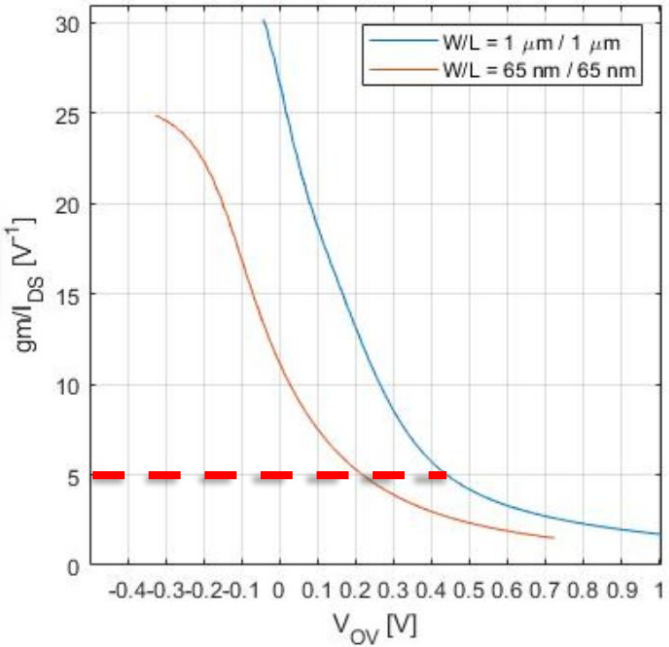
Exercise 1



f) Calculate, what $\frac{g_m}{I_D}$ is required to have exactly $A_V = 0$ and $V_{swing} = 0.8V_{pp}$.

Expression for gain

f) $$\frac{g_{m_{1,2}}}{I_{D_{1,2}}} = \frac{g_{m_{1,2}}}{(I_{SS}/2)} = \frac{g_{m_{1,2}} \cdot R}{(I_{SS} \cdot R/2)} = \frac{A_V}{\left(\frac{V_{swing}}{2}\right)/2} = \frac{A_V}{V_{swing}/4} = 4 \cdot \underbrace{\frac{A_V}{V_{swing}}}_{\frac{1}{0.8}} = 5$$



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Exercise 1

g) Calculate R and g_m such that this circuit passes exactly the V_{swing}-, P_{diss} and A_V-specification.

g) First, calculate the max. allowed current

$$P_{diss} = V_{DD} \cdot (I_{SS} + I_{bias}) = 1.2\ mW$$

$$\Leftrightarrow I_{SS} = \frac{P_{diss} - V_{DD} \cdot I_{bias}}{V_{DD}} \approx 1mA$$

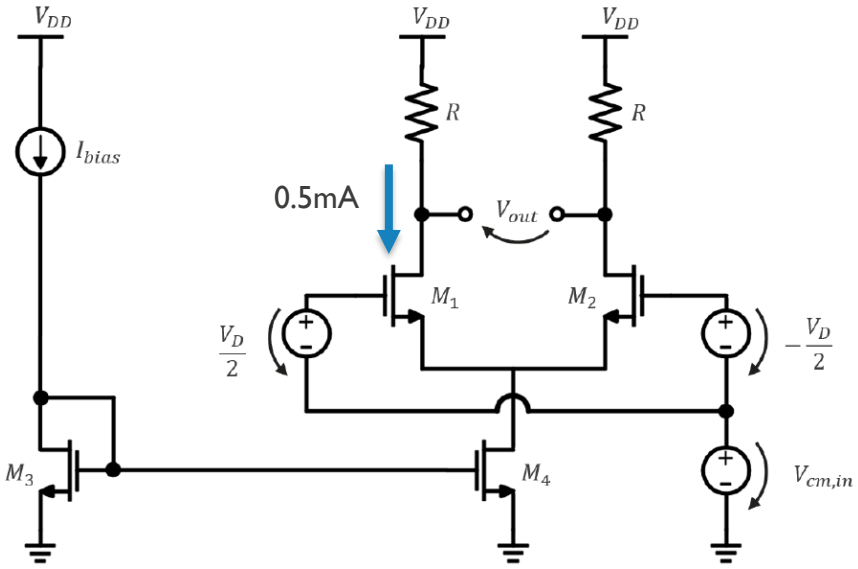
Then, calculate $g_{m_{1,2}}$ (from $\frac{g_{m_{1,2}}}{I_{D_{1,2}}}$ found previously)

$$g_{m_{1,2}} = \left(\frac{I_0}{2}\right) \cdot 5 = 2.5mS$$

Now, calculate R

$$A_{V,diff} = g_{m_{1,2}} \cdot R$$

$$\Leftrightarrow R = \frac{A_{V,diff}}{g_{m_{1,2}}} = 400\ \Omega$$



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Exercise 1

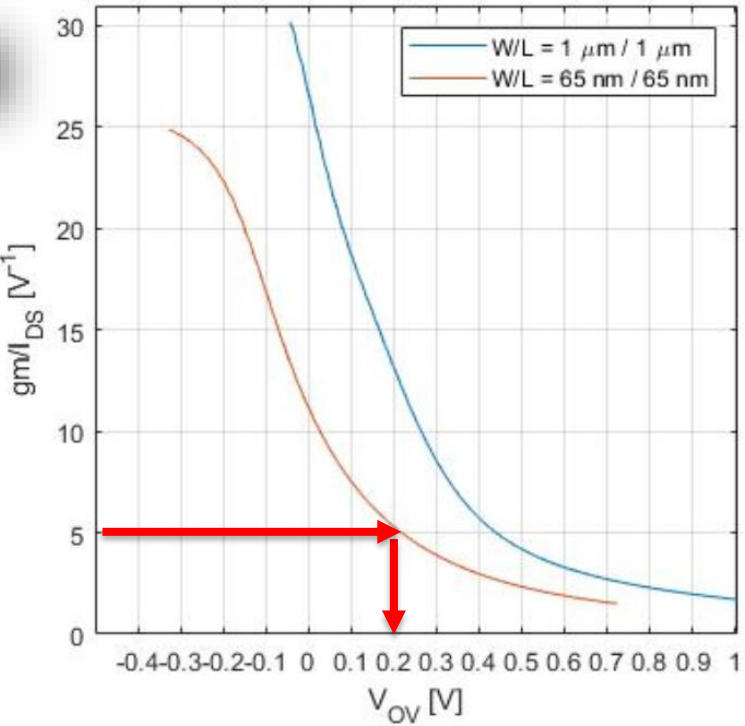
h) Calculate the required width W_1 and W_2 as well as L_1 and L_2 with aid of Fig. 5 and Fig. 6.

h) Reminder: $\frac{g_{m_{1,2}}}{I_{D_{1,2}}} = 5$

Set length to minimum, i.e. $L_{1,2} = 65nm$ $\left(f_T \sim \frac{1}{L^2}\right)$

Determine $V_{OV} \rightarrow 0.2V$

$$f_T = \frac{g_m}{2\pi C}$$



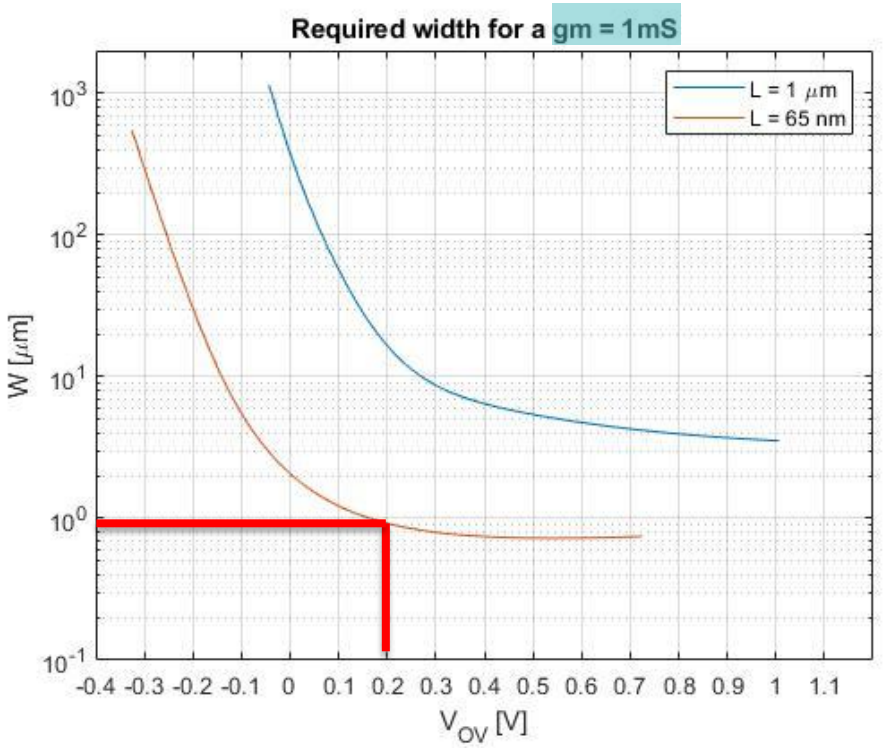
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Exercise 1

h) Now determine the width $W_{1,2}$ at $V_{OV} = 0.2V$

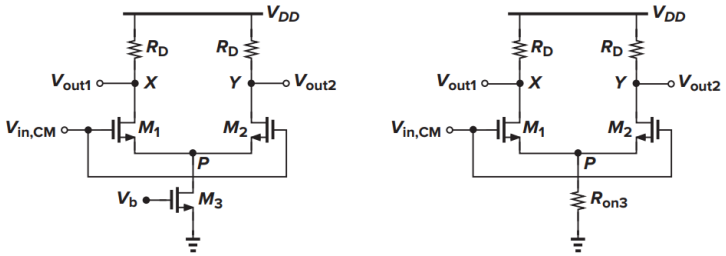
from g): $g_m = 2.5mS = 2.5 \cdot 1mS$

W $\Rightarrow 0.9\mu m \cdot 2.5 = 2.25\ \mu m$



Session 3 : Design of differential pairs and of an OTA

Common Mode Analysis



What happens if $V_{in,CM}$ varies from 0 → V_{DD}

$V_{in,CM} = 0$

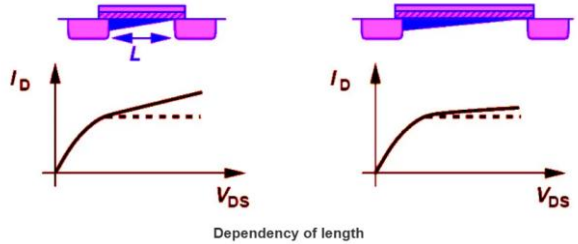
M1, M2 OFF → ID3 = 0

$V_{in,CM} > V_{th}$

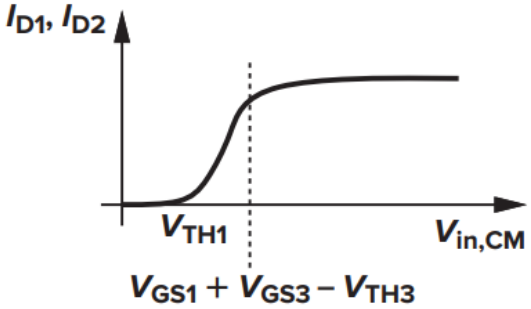
M1, M2 ON → ID3 increases

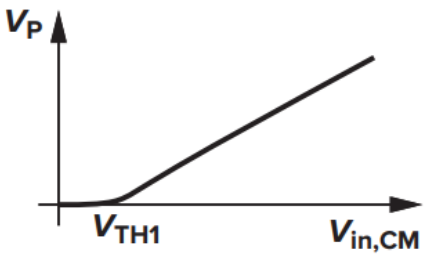
$V_{in,CM} > V_{GS1} + V_{GS3} - V_{th3}$ Saturation!

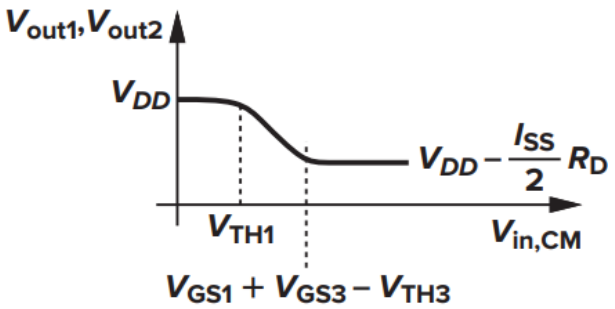
M1, M2 ON → ID3 constant



Dependency of length







Session 3 : Design of differential pairs and of an OTA

Exercise 1

i) Decide what length L_3, L_4 and V_{OV} you would choose for M_3 and M_4. Explain why? Then, calculate the width W_3 and W_4 with aid of Fig. 6.

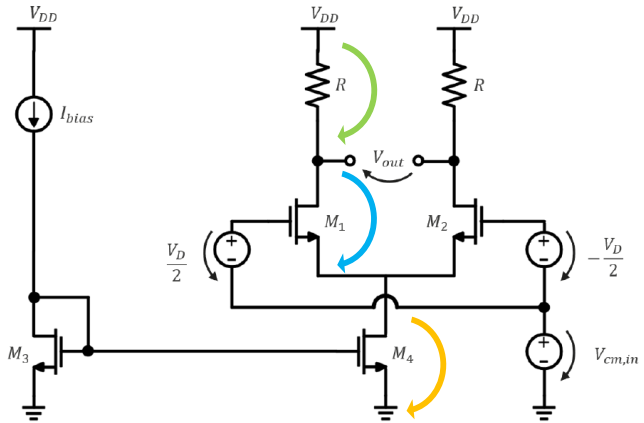
$V_{Dsat} \approx V_{OV}$ (very pessimistic assumption)

to ensure saturation for all M_x:

$$V_{OV,3,4} + V_{OV,1,2} + \frac{V_{swing}}{2} \leq V_{DD}$$

i) $L_{3,4} = 1\mu m$ to have negligible CLM (good current source)

Then, decide for a suitable value for $V_{OV_{3,4}} \rightarrow 0.3\ V$



Session 3 : Design of differential pairs and of an OTA

Exercise 1

Reminder $I_{SS} = 10 I_{bias}$

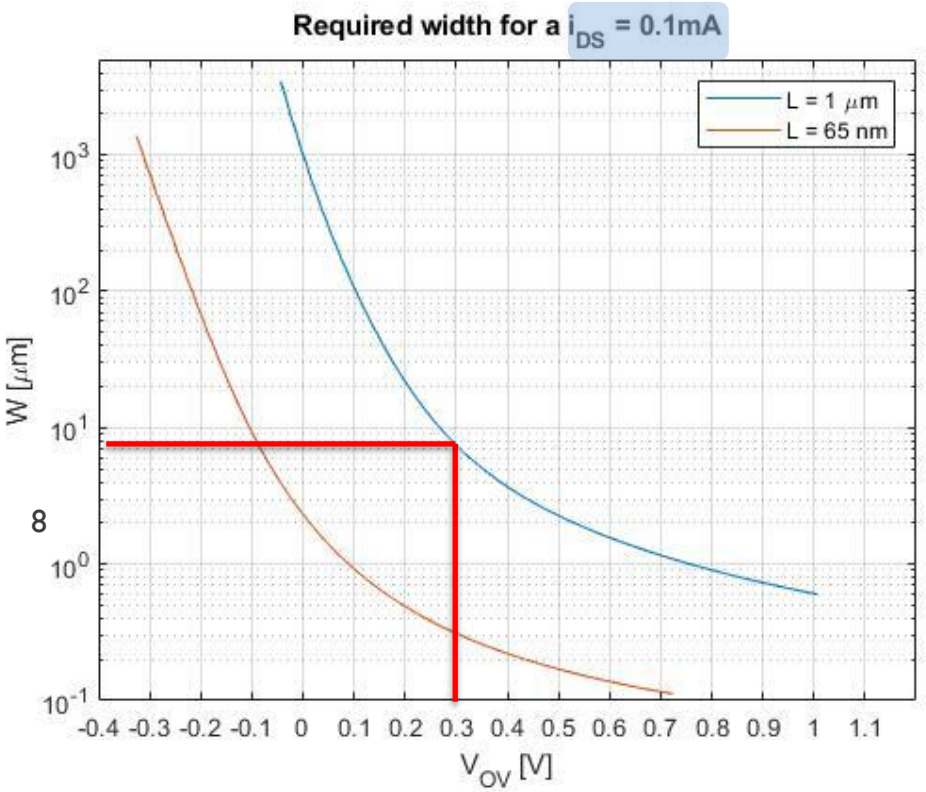
i) Calculate the width $W_{3,4}$

$$I_{DS,4} = I_{SS} = 1mA = 10 \cdot \overbrace{0.1mA}^{I_{bias}}$$

$$W_4 \Rightarrow 8\,\mu m \cdot 10 = 80\mu m$$

Then, with current mirror equation determine W_3

$$\frac{I_{SS}}{I_{bias}} = \frac{W_4}{W_3} \Leftrightarrow W_3 = \frac{W_4}{(I_{SS}/I_{bias})} = \frac{W_4}{10} = 8\mu m$$



Session 3 : Design of differential pairs and of an OTA

Exercise 1

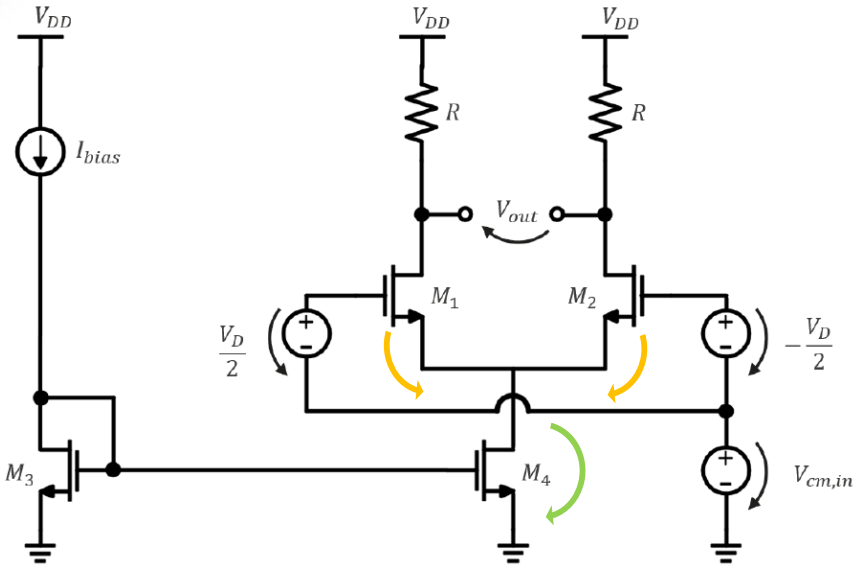
j) Calculate the minimum allowed input common mode voltage $V_{cm,min,in}$. What happens, if one sets $V_{cm,in} < V_{cm,min,in}$?

j) $V_{cm,in,min}$ calculated as follows

$$V_{cm,in,min} = V_{Dsat4} + V_{GS_{1,2}}$$

$$= V_{Dsat4} + V_{OV_{1,2}} + V_{th,n}$$

$$= 0.3V + 0.2V + 0.4V = 0.9V$$



Session 3 : Design of differential pairs and of an OTA

Exercise 1

j) What happens, if $V_{cm} < V_{cm,in,min}$?

→ drives M4 into triode ....

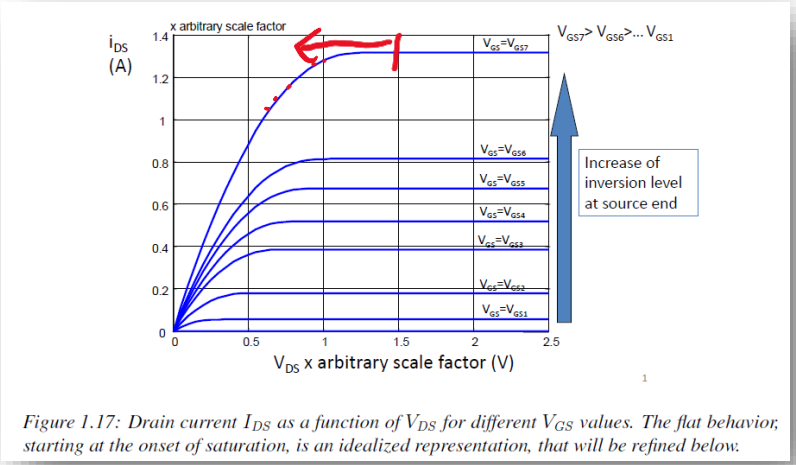
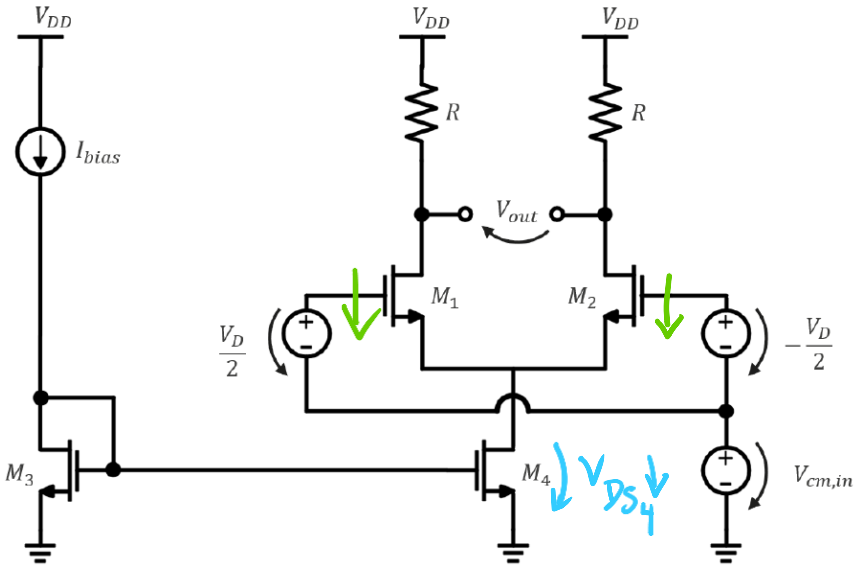


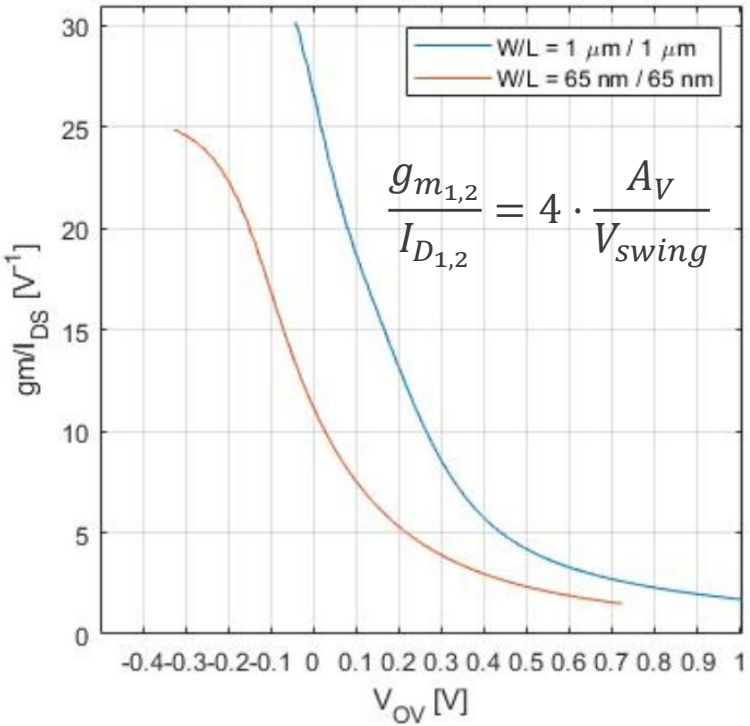
Figure 1.17: Drain current I_{DS} as a function of V_{DS} for different V_{GS} values. The flat behavior, starting at the onset of saturation, is an idealized representation, that will be refined below.



Session 3 : Design of differential pairs and of an OTA

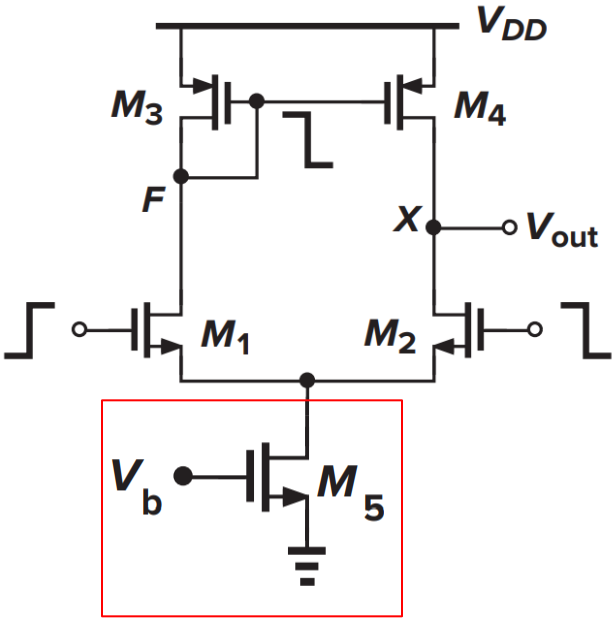
Summary

- Value the $\frac{g_m}{I_D}$-plot !
  - ➢ With $\frac{g_m}{I_D}$ one can identify the entire trade-off in one plot !



Session 3 : Design of differential pairs and of an OTA

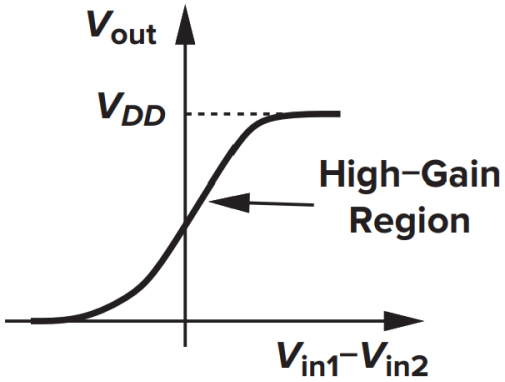
Differential Pair with Active Load



When the input at M1 increase, ID1 increases, VF falls, and ID2 decreases.

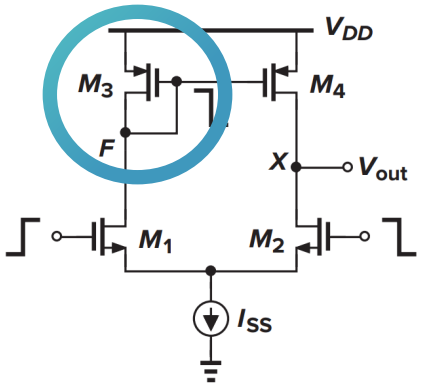
Thus, the output voltage rises by means of two mechanisms:

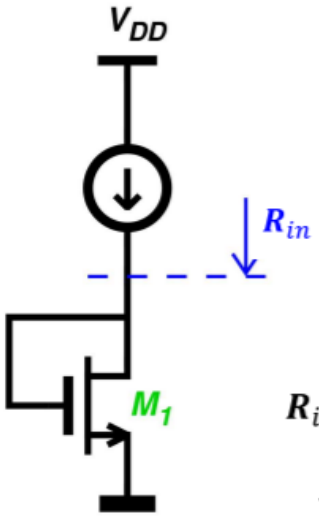
- M2 draws less current from X to ground and,
- M4 pushes a greater current from VDD to X.

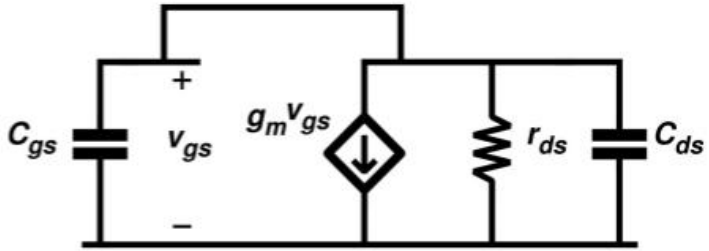


Session 3 : Design of differential pairs and of an OTA

Differential Pair with Active Load





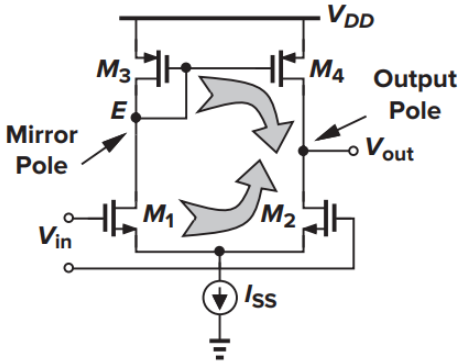


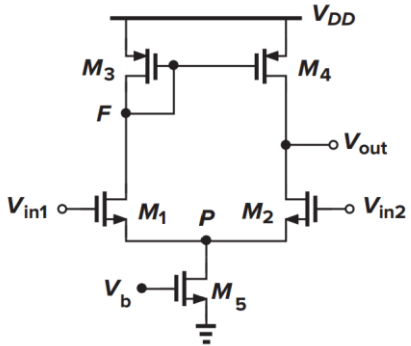
$$R_{in} = \frac{1}{g_m} || r_{ds} = \frac{1}{g_m + g_{ds}} \approx \frac{1}{g_m}$$

low impedance!

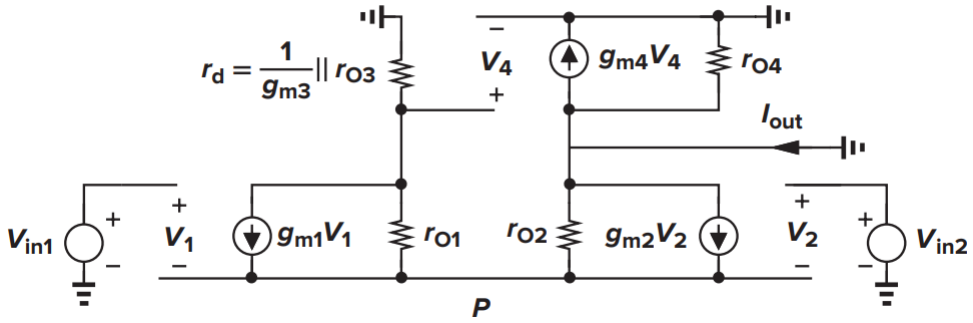
Session 3 : Design of differential pairs and of an OTA

Differential Pair with Active Load









Small signal equivalent

Hint!

Session 3 : Design of differential pairs and of an OTA

- Solve exercise 2 a-e
  - ETA: 30 min

Session 3 : Design of differential pairs and of an OTA

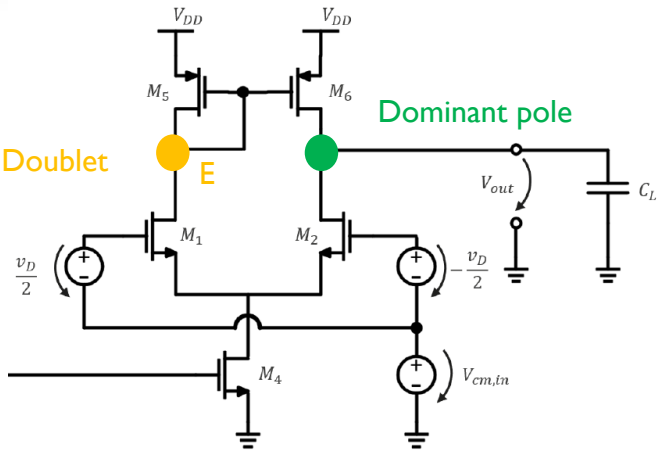
Exercise 2

a) Give the small signal voltage gain equation of this circuit including the capacitances and indicate

(i) dominant pole,

(ii) non-dominant pole,

(iii) zero.



Dominant pole $\omega_{p1} \approx \frac{1}{(r_{ON} \| r_{OP}) C_L}$

Non-dominant Pole $\omega_{p2} \approx \frac{g_{mP}}{C_E}$

Zero $\omega_{p2} \approx \frac{g_{mP}}{C_E} \times 2$

Session 3 : Design of differential pairs and of an OTA

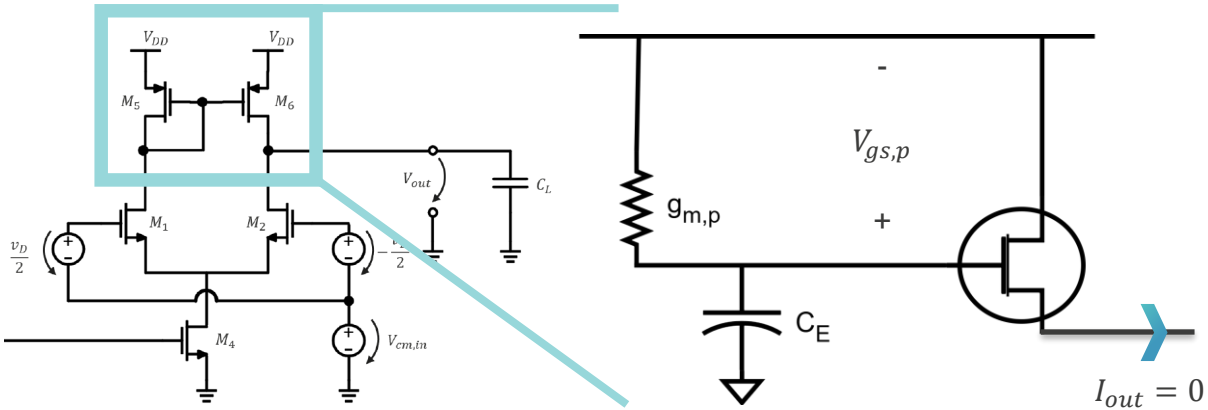
Exercise 2: how to find the zero?

a) Give the small signal voltage gain equation of this circuit including the capacitances and indicate

(i) dominant pole,

(ii) non-dominant pole,

(iii) zero.



Equate the currents → $-V_{gs,p}g_{mp} = \frac{V_{gs,p}}{\frac{1}{g_{mp}} \| \frac{1}{sC_E}}$

$$-g_{mp} = g_{mp} + sC_E$$

$$s_z = \frac{-2g_{mp}}{C_E}$$

Circuit consists of a "slow path" (M1, M5, and M6) in parallel with a "fast path" (M1 and M2).

Session 3 : Design of differential pairs and of an OTA

Exercise 2

a) Give the small signal voltage gain equation of this circuit including the capacitances and indicate

(i) dominant pole,

(ii) non-dominant pole,

(iii) zero.

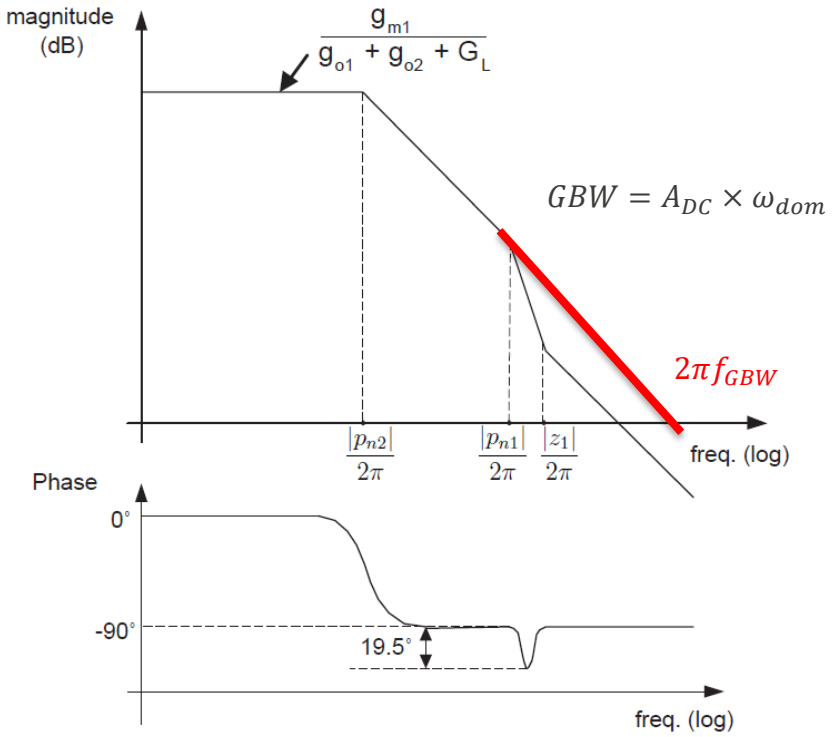
b) Assume the pole-zero-doublet to be far away from the dominant pole. Draw the bode- and phase plot and indicate location of the dominant pole and GBW.

a)

$$A_v(s) = \frac{v_{out(s)}}{v_{in(s)}} = \frac{g_{m,1,2}}{g_{02}+g_{06}} \cdot \frac{1 + s\frac{C_{d1,5}}{2gm_5}}{1 + s\frac{C_{d1,5}}{gm_5}} \cdot \frac{1}{1 + s\frac{C_{d2,6}}{g_{02}+g_{06}}}$$

zero — non-dominant pole — dominant pole

b) see drawing



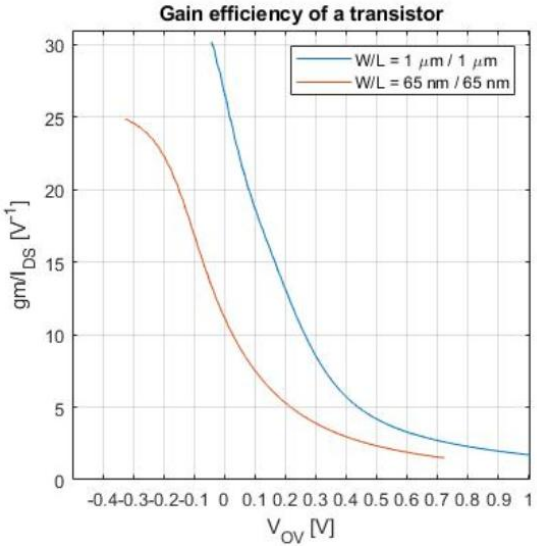
Session 3 : Design of differential pairs and of an OTA

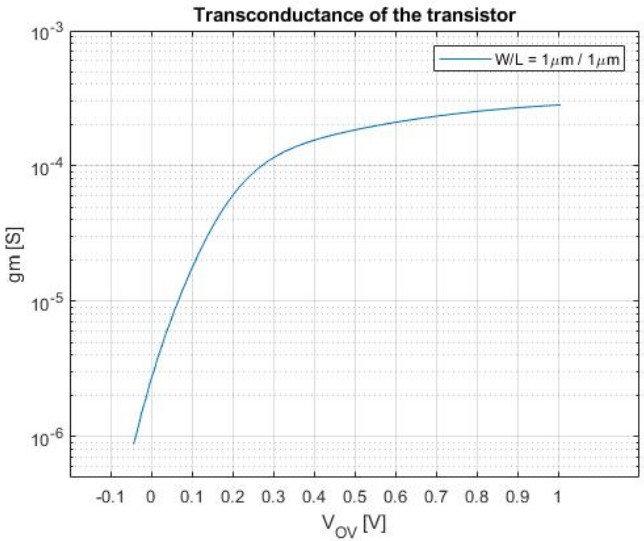
Exercise 2

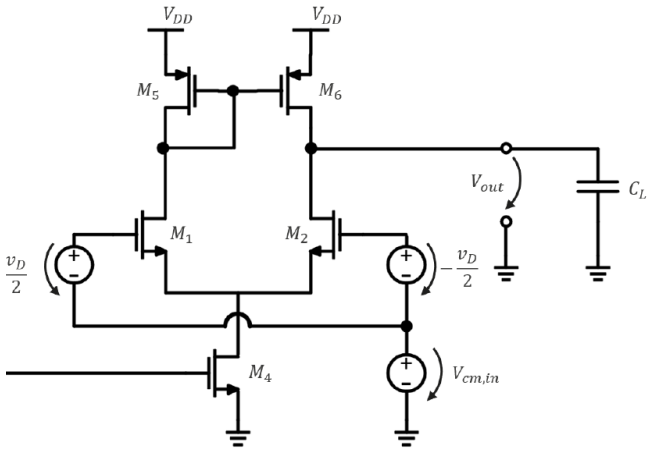
c) What happens to the pole-zero-doublet, if one increases/decreases the V_{OV} of M_5 and M_6 assuming that the current flowing through M4 is fixed as well as the lengths are fixed?

c) $p_{non-dom} \sim \frac{g_{m5}}{C_{gs_{5,6}}}$

$V_{OV,5} \uparrow \Rightarrow g_{m5}\ ? \Rightarrow p_{nom-dom}\ ?$







Session 3 : Design of differential pairs and of an OTA

Exercise 2

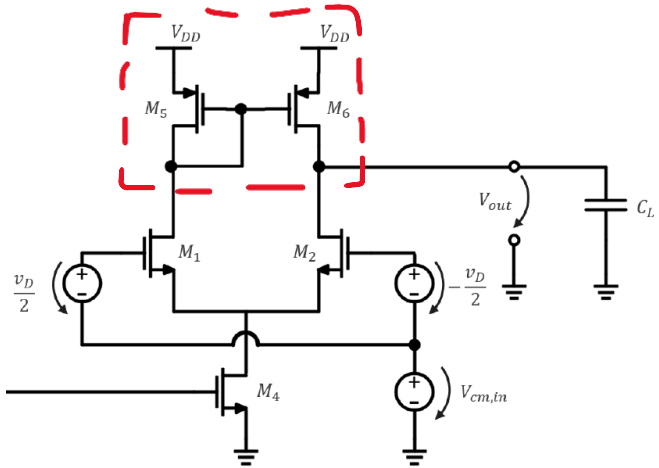
c) $p_{non-dom} \sim \frac{g_{m5}}{C_{gs_{5,6}}} \sim \frac{2 \cdot I_{DS,5}}{V_{OV_5} \cdot W_5} \sim V_{OV_5}$

$C_{gs_{5,6}} \sim C_{ox} \cdot W \cdot L$

$W_5 \sim \frac{2I_{DS,5}}{V_{OV,5}^2}$

$I_{DS} = \mu C'_{ox} \cdot \frac{W}{2L} \cdot V_{OV}^2$

$V_{OV,5} \uparrow \Rightarrow p_{nom-dom} \uparrow$



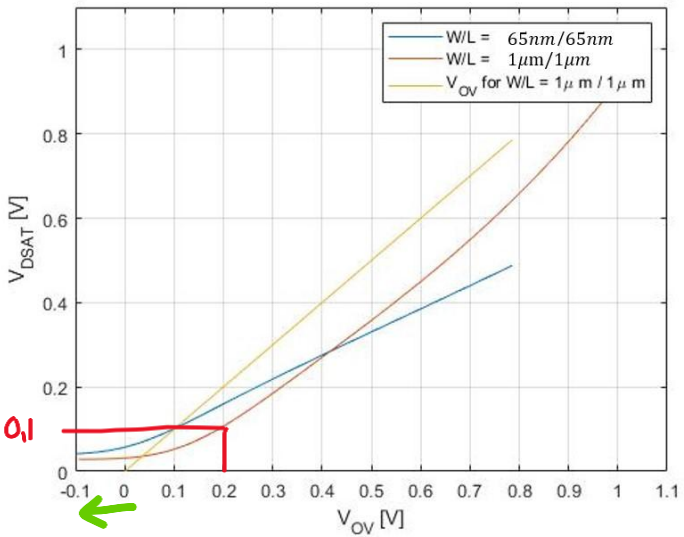
Session 3 : Design of differential pairs and of an OTA

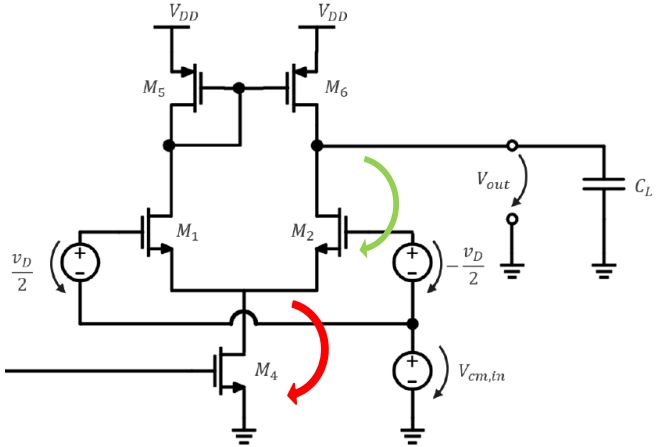
Exercise 2

d) Assume $V_{OV,3,4} = 0.2V$, $V_{Dsat,5,6} = -0.25V$ and $V_{OV} < -0.1V$ for the remaining devices. With aid of Fig. 9, calculate the maximum/minimum possible output common mode voltage $V_{cm,max,out}$ and $V_{cm,min,out}$, where all transistors are still in saturation.

d) Calculate $V_{out,cm,min}$ (for $V_{OV,4} = 0.2V$ and remaining $V_{OV} < 0V$)

$$V_{out,cm,min} = V_{Dsat,4} + V_{Dsat,1,2}$$
$$= 0.1\,V + 0.05V = 0.15\,V$$





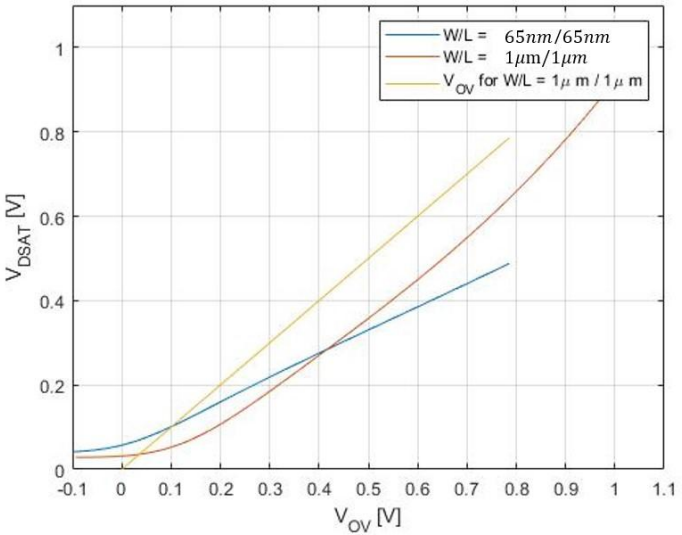
Session 3 : Design of differential pairs and of an OTA

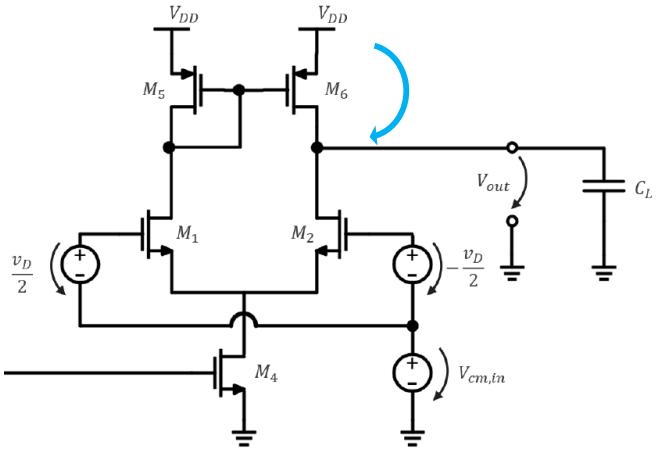
Exercise 2

d) Calculate $V_{out,cm,max}$ (for $V_{OV,4} = 0.2V$ and $V_{Dsat,5,6} = -0.25\text{V}$)

$$V_{out,cm,max} = V_{DD} + V_{Dsat,5,6}$$

$$= 1.1\,V - 0.25V = 0.85\,V$$





Session 3 : Design of differential pairs and of an OTA

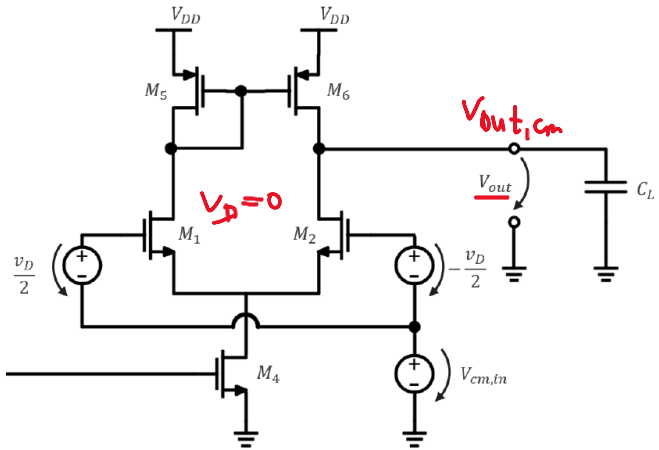
Exercise 2

e) The designer decides to set the output common mode voltage for $v_D = 0$ to the midpoint between $V_{cm,max,out}$ and $V_{cm,min,out}$. Explain how and calculate for that the required V_{OV}.
Hint: Examine the PMOS pair for $v_D = 0$.

e) $V_{out,cm}$ calculated as follows

$$V_{out,cm} = \frac{V_{out,cm,max} + V_{out,cm,min}}{2} = 0.5V$$

How to setup $V_{out,cm}$ in OTA's ?



Session 3 : Design of differential pairs and of an OTA

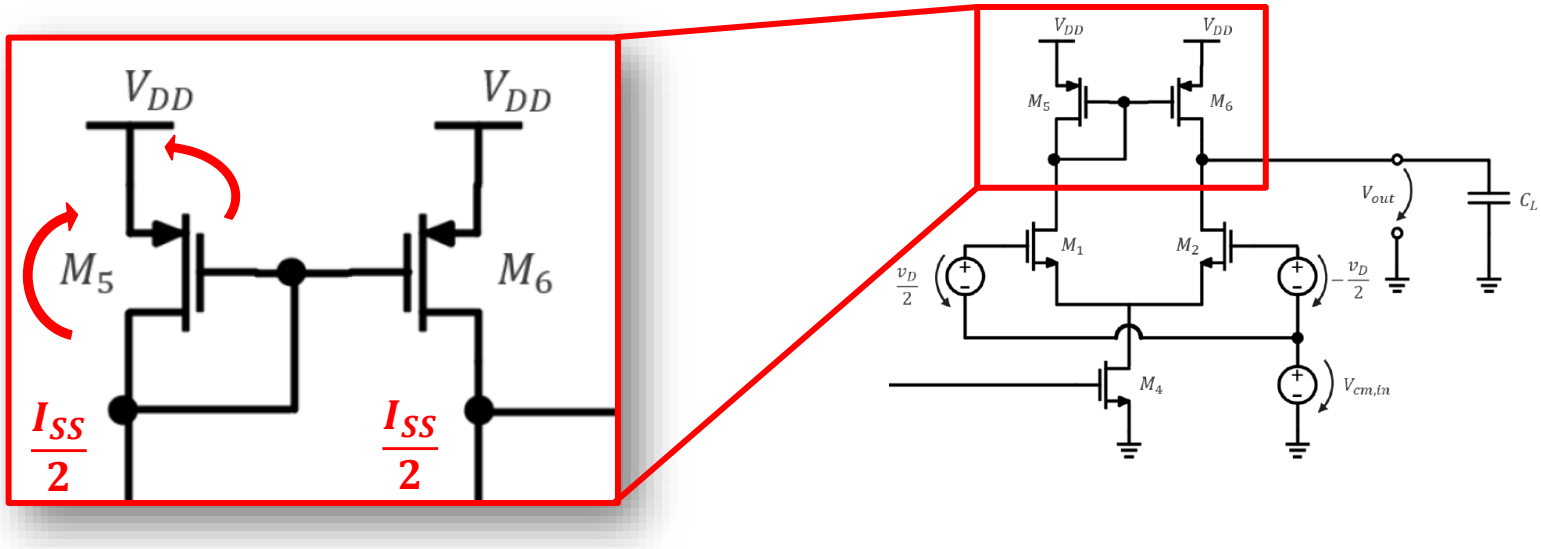
Exercise 2

e) $V_{out,cm} = \frac{V_{out,cm,max} + V_{out,cm,min}}{2} = 0.5V$

How to setup $V_{out,cm}$ in OTA's ?

$V_{DS5} = V_{GS5}$

$V_{DS5} = V_{GS5} = V_{DS6}$

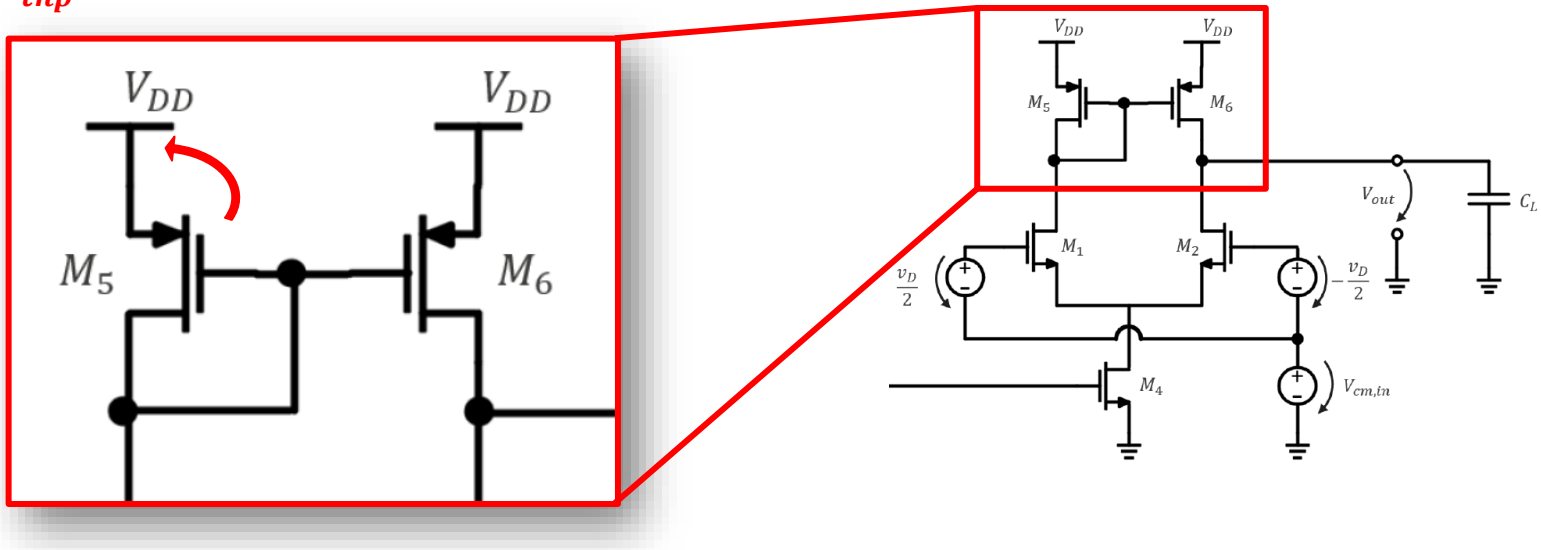


Session 3 : Design of differential pairs and of an OTA

Exercise 2

e) Consequently, $|V_{OV,5}| = V_{DD} - V_{out,cm} - |V_{th,p}| = 0.38V$

$\boldsymbol{V_{GS5} = V_{OV6} + V_{thp}}$



Session 3 : Design of differential pairs and of an OTA

- Solve exercise 2 f-i
  - ETA: 20 min

Session 3 : Design of differential pairs and of an OTA

Exercise 2

f) What is the expression for A_v in low frequencies (below the dominant pole frequency)? Use Fig. 8 to calculate required V_{OV} and $V_{in,cm,min}$ for M1 and M2 to achieve $A_v > 8$.

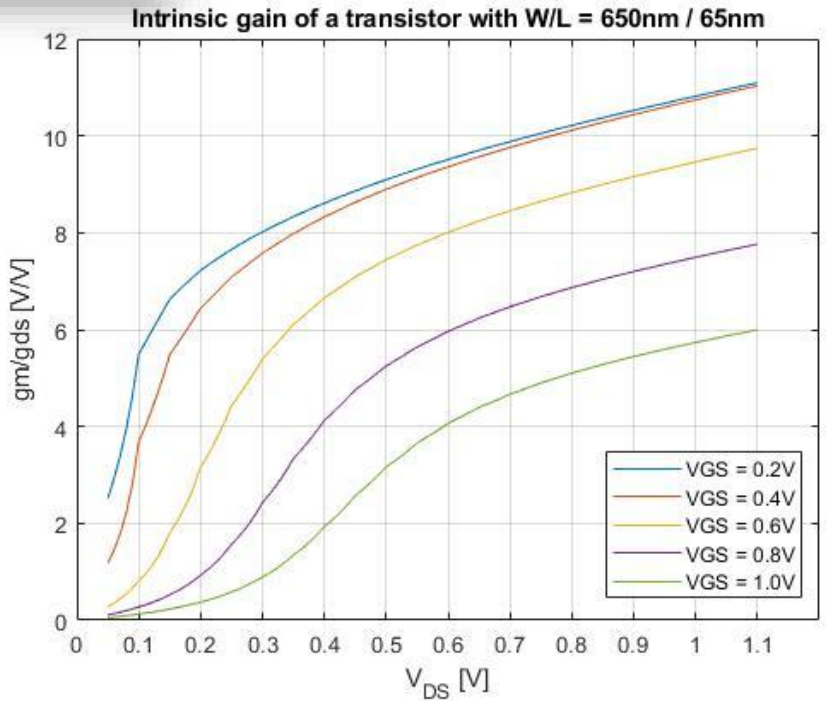
f) How to simplify $A_V(s = 0)$?

$$A_v(s) = \frac{v_{out(s)}}{v_{in(s)}} = \frac{g_{m,1,2}}{g_{02} + g_{06}} \cdot \frac{1 + s\frac{C_{d1,5}}{2g_{m5}}}{1 + s\frac{C_{d1,5}}{g_{m5}}} \cdot \frac{1}{1 + s\frac{C_{d2,6}}{g_{02}+g_{06}}}$$

zero — non-dominant pole — dominant pole

⇒ make $L_{5,6} = 1\mu m$ for negligible CLM
⇒ $g_{02} \gg g_{06}$

$$A_V(s = 0) \approx \frac{g_{m,1,2}}{g_{02}}$$

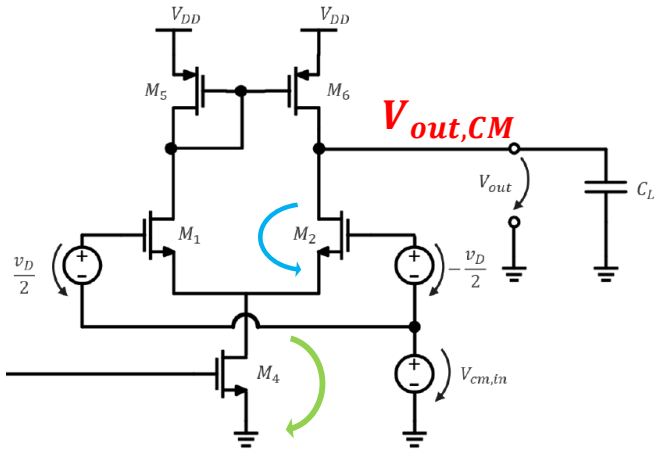


Session 3 : Design of differential pairs and of an OTA

Exercise 2

f) $V_{DS,2}$ calculated as follows

$$V_{DS,2} = V_{out,cm} - V_{Dsat,4} = 0.5 \ - 0.1V = 0.4V$$



Session 3 : Design of differential pairs and of an OTA

Exercise 2

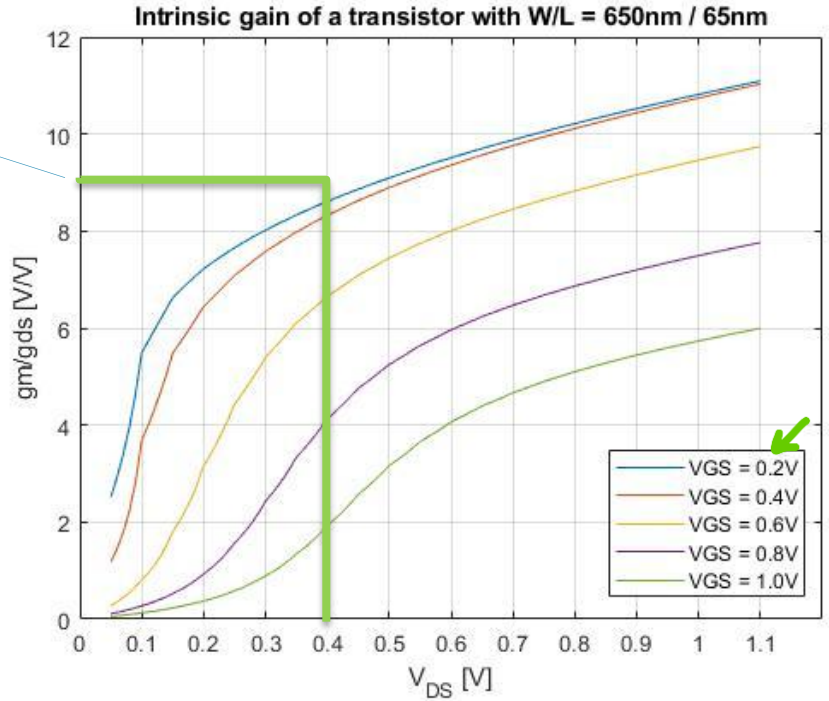
f) Search $V_{OV,1,2}$ for $A_V(s=0) > 8$

$\rightarrow V_{GS,1,2} = 0.2V \rightarrow A_V \sim 9$

$\rightarrow V_{OV,1,2} = V_{GS,1,2} - V_{th,n} = -0.2V$

0.4 V

$\rightarrow V_{in,cm} = V_{GS,1,2} + V_{Dsat,4} = 0.3V$



Session 3 : Design of differential pairs and of an OTA

Exercise 2

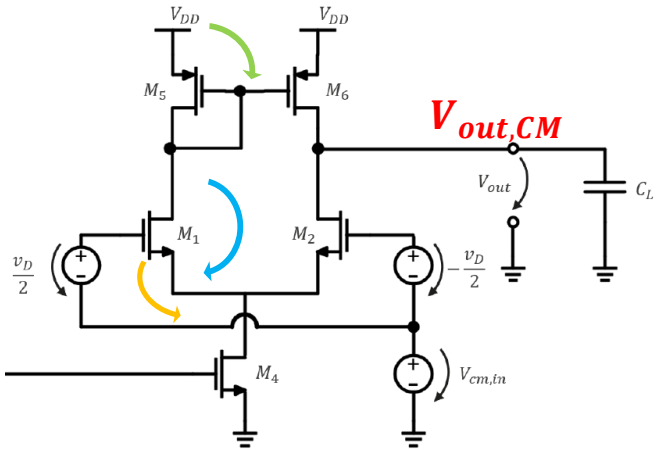
g) Calculate $V_{in,cm,max}$.

g) $V_{in,cm,max}$ calculated as follows:

$$V_{in,cm,max} = V_{DD} + V_{GS,5} - V_{DSat1,2} + V_{GS,1,2}$$

$$= V_{out,cm} - V_{Dsat,1,2} + V_{GS,1,2}$$

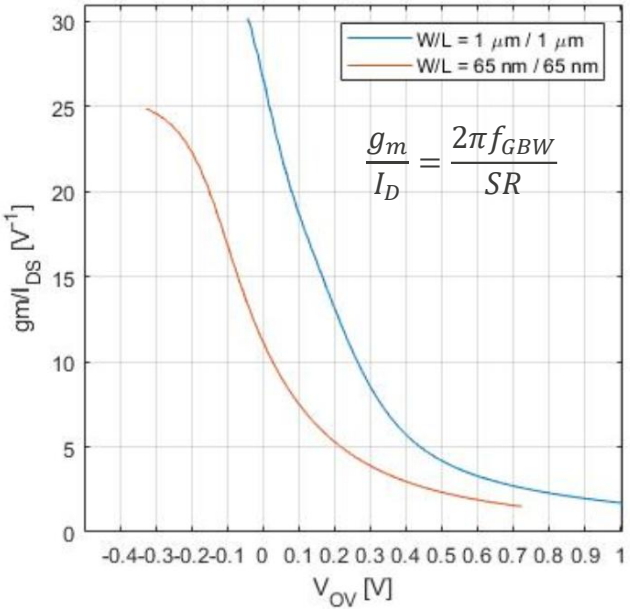
$$= \underbrace{0.5V}_{V_{out,CM}} - 0.05V + 0.2V = 0.65V$$

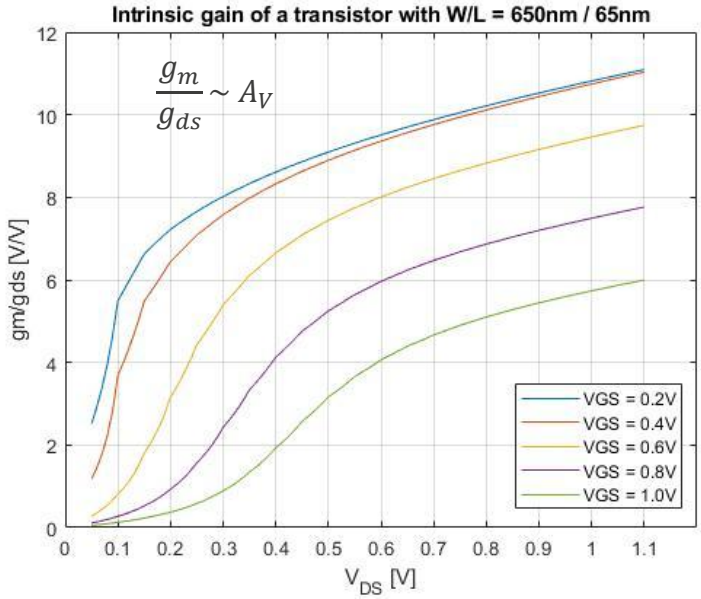


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Summary (1)

1. Value the $\frac{g_m}{I_D}$- and $\frac{g_m}{g_{ds}}$- plots!
   - ➢ One can predict the feasibility of the given specifications for an OTA in one view

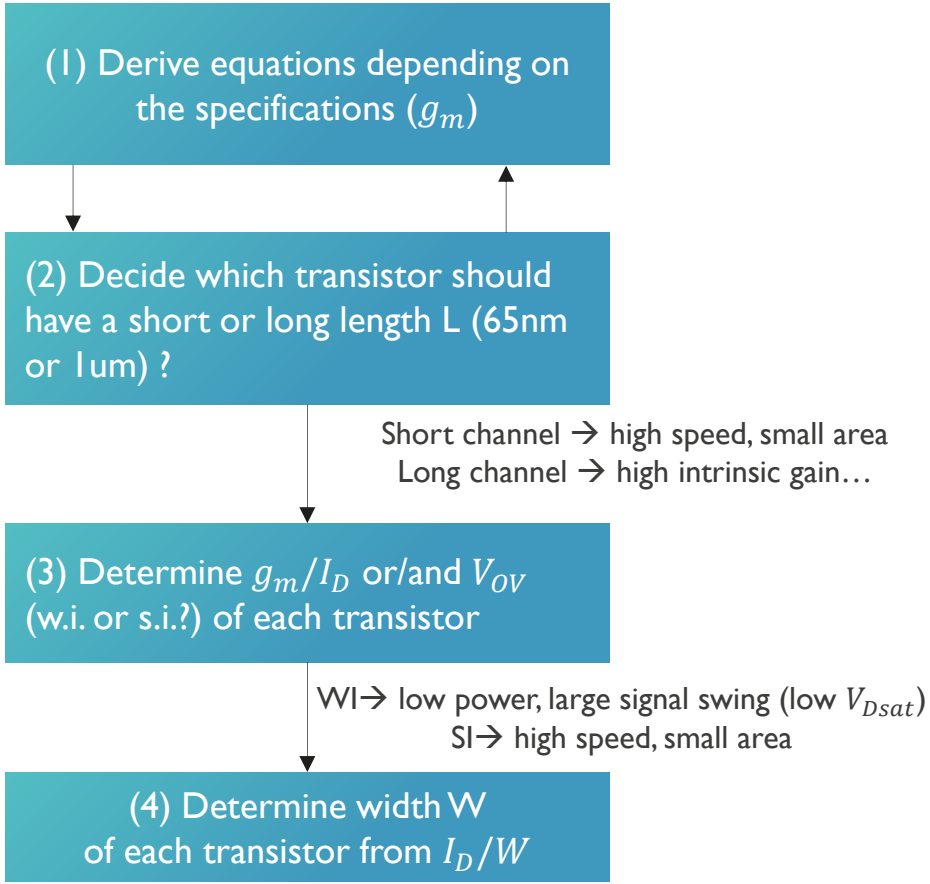




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Summary (2)

2. Design flow introduced
3. Learned how to "design on paper" with
   - W-plot for fixed g_m or i_{ds}
   - $\frac{g_m}{I_D}$- and $\frac{g_m}{g_{ds}}$-plots
   - V_{dsat}-plot
   - ➢ Those plots will help you to save time when designing for the final lab session!



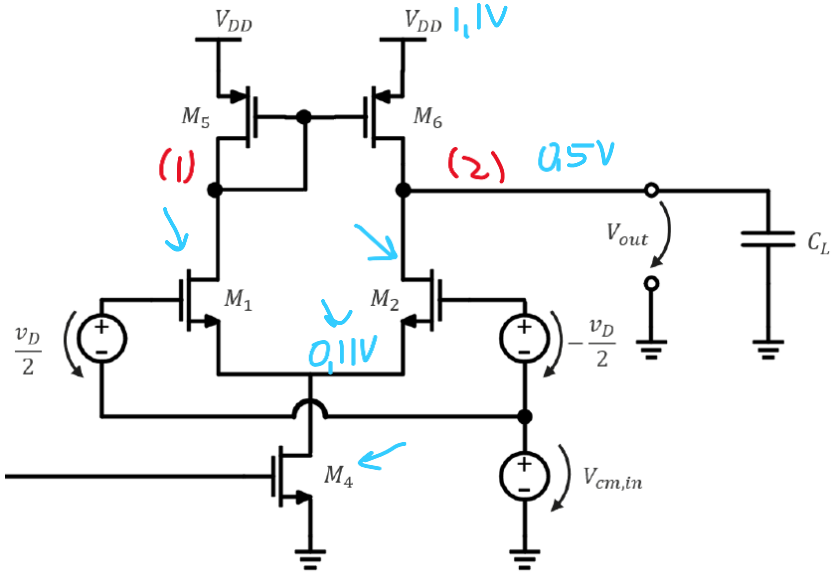
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MATLAB session

- Exercise 3
  1. Type in the code accordingly to this presented OTA exercise
  2. Solve following scientific questions?
     1. How can one move the pole-zero-doublet?
        - 2 options
     2. How can I increase the f_{GBW}?
  - ETA: 30 min

Plug in the values below!

```
M4.ids    = 200e-6:
SR          = M4.ids/spec.CL/1e6 % [mV/ns] slew rate
```



Hints:

1. Use the following command to copy transistors e.g. M1 and M2
   1. M2 = cirElementCopy(M1, M2);
2. Before coding, draw all voltage condition on the OTA-circuit
   - i.e. vgs, vgd, vds, ids (or gm) of each transistor !
   - This will help you to speed-up your coding
3. Code it smart !
   1. code it such that you can play around with the pole-zero doublet !

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... some helpful references

[1]: “Analog Design Essentials”, Sansen, Willy M. C, 2006

[2]: “Design of Analog CMOS Integrated Circuits”, Razavi, Behzad, 2016

[3]: “Analysis and Design of Analog Integrated Circuits”, P. Gray, P. Hurst, S. Lewis and R. Meyer, John Wiley and Sons, 2009



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